

JUHO JEON

South Korea | +82 10-3956-3595 | ijeonaa@connect.ust.hk | jeon-juho.github.io

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Sep 2024 – Dec 2026 (Expected)

Master of Philosophy (MPhil) in Electronic and Computer Engineering

Hong Kong

- GPA: 3.717 / 4.30 | Advisor: Prof. Patrick Chik Yue
- Teaching Assistant, ELEC3400 – Integrated Circuit Designs and Systems (Prof. Howard Luong, Spring 2025)
- Relevant coursework: Advanced Analog IC Analysis and Design; Power Management Integrated Circuit Design; RF Microsystems: Devices and Applications; Advanced AI Chip and System; High Frequency Circuit Design; Electronic Design Automation for VLSI Design.

The Hong Kong University of Science and Technology (HKUST)

Sep 2017 – Jun 2024

Bachelor of Engineering in Electronic Engineering; Minor in Information Technology

Hong Kong

- GGA: 3.239 / 4.30 | Second Class Honours, Division I
- Academic leave: Sep 2019 – Aug 2021 (mandatory military service, Republic of Korea); Feb 2022 – Feb 2023 (full-time internship, Deloitte Advisory)
- Relevant coursework: Analogue Integrated Circuits Design and Analysis; Integrated Circuits and Systems; CMOS VLSI Design; Integrated Power Electronics; Electronic Circuits; Signals and Systems; Signal Processing and Communications; Probability and Random Processes in Engineering; FPGA-based Design; Embedded Systems.

PUBLICATIONS

- J. Jeon and C. P. Yue, "Digitally Reconfigurable Resolution Multi-Bit-per-Cycle SAR ADC: Design Trade-off Considerations," **IEEE Region 10 Conference (TENCON) 2026**, accepted.

CIRCUIT DESIGN EXPERIENCE

28nm CMOS Reconfigurable Multi-Resolution SAR ADC — MPhil Thesis

Sep 2024 – Present

Advisor: Prof. Patrick Chik Yue | Post-Layout Simulation

- Designed a digitally reconfigurable SAR ADC supporting five resolution modes (5b–9b) via a 1-then-2-bit-per-cycle conversion scheme, with reconfigurability implemented entirely through digital logic — no switches added to the analog signal path.
- Achieved 51.84 dB SNDR, 8.31 ENOB, and 66.08 dB SFDR at 1.4 GS/s (0.9 V supply, 9b mode), corresponding to a 23.8 fJ/conversion-step Walden FoM and 160 dB Schreier FoM, verified using post-layout extracted netlists for the DAC and comparator.
- Designed an enable-gated DAC switching architecture and mode-aware decoder allowing seamless mode transitions via a 4-bit control word with no analog reconfiguration.
- Quantified the high-resolution-limited power trade-off ($\sim 9\times$ FoM degradation from 9b to 5b mode) arising from sharing a single DAC/comparator across all resolution modes.
- Results accepted for presentation at IEEE TENCON 2026.

180nm CMOS 10-bit SAR ADC — Final Year Thesis

Sep 2023 – May 2024

Advisor: Prof. Zhang Yihan | Schematic-Level Simulation, TSMC 180nm | First Runner-Up, ECE Best FYP Award

- Designed and independently verified a 10-bit monotonic-switching SAR ADC (capacitive DAC, bootstrapped sampling switch, dynamic comparator, asynchronous SAR logic) in Cadence Virtuoso, achieving 9.09 ENOB and 1.92 mW power consumption at 2 MS/s.

28GHz Differential LNA Design

Spring 2026

RFIC Design Coursework (Prof. Patrick Chik Yue) | 22nm TSMC CMOS

- Designed a 28GHz differential cascode LNA with inductive source/gate degeneration, achieving 17.4 dB gain, 1.74 dB NF, -14.1 dB S11, and 9.16 dBm IIP3 at 22.8 mW.

Gain-Boosted Folded-Cascode Fully-Differential OTA

Fall 2024

Advanced Analog IC Design Coursework (Prof. Howard Luong) | 90nm TSMC CMOS

- Designed a gain-boosted folded-cascode OTA with resistive CMFB, achieving 81.6 dB gain, 604 MHz UGF, 61.5° phase margin, 103 dB CMRR, and -66 dB THD at 1.93 mW (± 1 V supply).

Continuous-Time Gm-C Channel-Selection Filter for UWB

Fall 2024

Advanced Analog IC Design Coursework (Prof. Howard Luong) | 90nm TSMC CMOS

- Designed a Gm-C UWB bandpass filter (35 kHz–259 MHz) from passive Chebyshev-I prototypes through transistor-level implementation, meeting passband gain/ripple, corner-frequency, and adjacent-channel specs.

DC-DC Buck Converter with PWM Voltage-Mode Control

Fall 2024

Power Management IC Design Coursework (Prof. Wing Hung Ki) | 180nm TSMC CMOS

- Designed a 3.3V-to-1.8V, 1MHz synchronous buck converter achieving 86.2% peak efficiency and 45.2 mV ripple across 200–800 mA CCM load.

WORK EXPERIENCE

Deloitte Advisory

Feb 2022 – Feb 2023

Cyber Security Intern

Hong Kong

- Built and deployed identity and access management (IAM) workflows and a demonstration web application (Okta, Salesforce, Hong Kong iAMSmart) for client engagements.

Gense Technologies

Dec 2021 – Jan 2022

Electronic Engineer Intern

Hong Kong

- Verified and validated a portable medical imaging device through testbench characterization and clinical trial, confirming compliance with design specifications.
- Proposed an alternative constant current source design to meet a lower load-regulation requirement.

Republic of Korea Army

Sep 2019 – Apr 2021

Sergeant, Wireline Communication Squad Leader

Republic of Korea

- Led a wireline communication squad as squad leader, directing installation and maintenance operations.
- Installed, maintained, and secured military communication systems including wireline and optical wireline transceivers, satellite communication systems, wireless transceivers, alarm systems, and surveillance cameras.

LEADERSHIP EXPERIENCE

Deputy Teaching Assistant Coordinator

Sep 2025 – Jul 2026

Electronic and Computer Engineering Dept., HKUST

- Planned and organized departmental programs and events for graduate teaching assistants; provided peer mentoring.
- Hosted the monthly ECE Future Leaders PG Seminar Series, a platform for graduate students to broaden knowledge within and beyond their fields.
- Guided postgraduate program outreach activities.
- Provided teaching materials for teaching assistants: Prevention and Guidelines for Plagiarism in Class.

ECE Student Ambassador

Sep 2023 – Jun 2024

Electronic and Computer Engineering Dept., HKUST

- Advised junior students on academic and non-academic matters; supported departmental outreach activities.
- Hosted and managed department-level gathering activities.

HONORS AND AWARDS

First Runner-Up — ECE Best Final Year Project/Thesis Award

Jun 2024

Electronic and Computer Engineering Dept., HKUST

Dean's List

Spring 2023

Electronic and Computer Engineering Dept., HKUST

Top 5 Innovative Solution — Sustainability Smart Campus Competition

Nov 2021

The Hong Kong University of Science and Technology

SKILLS AND LANGUAGES

Circuit Design: Analog/Mixed-Signal CMOS IC Design, SAR ADC & Data Converters, mm-Wave RF Circuits (LNA), Power Management ICs, PCB Design, FPGA, RTL-to-GDSII Flow, Digital Circuit Design

Tools: Cadence Virtuoso, Spectre, ADE, Synopsys, HSPICE, Altium Designer, Advanced Design System (ADS), Ansys HFSS, Xilinx Vivado

Programming: MATLAB, Python, Verilog, Verilog-A

Languages: Korean (Native), English (Professional), Mandarin Chinese (Intermediate)